

Rewriting History: A Recipe for Interventional Analyses to Study Data Effects on Model Behavior

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Abstract

We present an experimental recipe for studying the relationship between training data and language model (LM) behavior. We outline steps for intervening on data batches – i.e., “rewriting history” – and then retraining model checkpoints over that data to test hypotheses relating data to behavior. Our recipe breaks down such an intervention into stages that include selecting evaluation items from a benchmark that measures model behavior, matching relevant documents to those items, and modifying those documents before retraining and measuring the effects. We demonstrate the utility of our recipe through case studies on factual knowledge acquisition in LMs, using both cooccurrence statistics and information retrieval methods to identify documents that might contribute to knowledge learning. Our results supplement past observational analyses that link cooccurrence to model behavior, while demonstrating that extant methods for identifying relevant training documents do not fully explain an LM’s ability to correctly answer knowledge questions. Overall, we outline a recipe that researchers can follow to test further hypotheses about how training data affects model behavior. Our code is made publicly available to promote future work.¹

1 Introduction

When training LMs, the choice of pretraining data is generally believed to have a profound effect on what the learned models can do and how well they can do it. Numerous prior works have curated large pretraining datasets and studied the effect of pretraining data composition on downstream tasks (Li et al., 2024; Longpre et al., 2024), with

some explicitly predicting downstream task performance (Grattafiori et al., 2024; Bhagia et al., 2025; Magnusson et al., 2025). The primary motivation here has generally been performance improvement, not necessarily understanding model behavior, while mainly operating on the scale of entire pretraining datasets.

An equally important body of work relates pretraining data to model behavior to understand *how* data affects performance without necessarily optimizing for it, retraining over modified data to do so. This includes various data-behavior relationships, such as how relative frequencies of verb tenses change subject-verb agreement performance (Wei et al., 2021) or how pronouns affect gender bias (Biderman et al., 2023). Some recent work focuses on the specific behavior of factual knowledge acquisition, retraining models to understand how data gives rise to knowledge learning. This work operates across a range of scales, from pretraining from scratch on fictional biographies (Allen-Zhu and Li, 2024, 2025; Zucchet et al., 2025), to continued pretraining of a base model on carefully-constructed documents expressing factual relations (Zhang et al., 2024), and even intervening on intermediate stages of pretraining by injecting synthetic data to study the effect of repetition and paraphrasing (Chang et al., 2024) or entropy across activations (Kim et al., 2025) on fact learning and forgetting.

While the aforementioned works all modify pretraining data to study model behavior, each takes a specific approach that does not easily transfer to other hypotheses. Seeking to draw connections across these works and formalize their methodologies, we propose a general recipe for *interventional analyses* to study the relationship between pretraining data and model behavior at a fine-grained level through *retraining* on modified data. Our proposed recipe (illustrated in Fig. 1) consists of three stages: (1) selecting which items

¹<https://github.com/rahuln/pretrain-intervention>.

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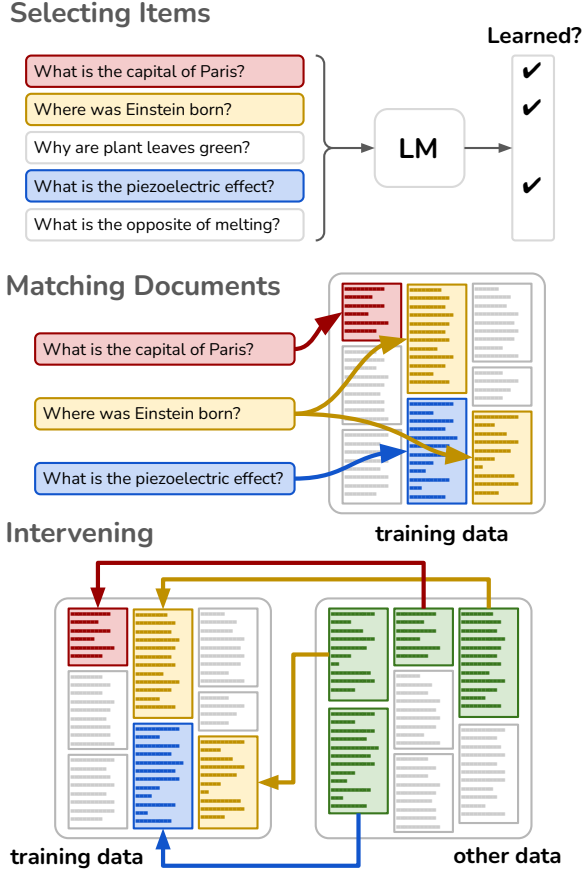


Figure 1: Diagram illustrating our proposed interventional analysis recipe, with stages including selecting items, matching documents to items, and intervening on matched documents.

in a benchmark to intervene on; (2) matching pretraining documents to those items; and (3) modifying those documents before repeating that portion of pretraining and measuring the effects. Our recipe organizes and comprehensively articulates the different considerations one must take into account when performing such an intervention, contextualized within past work. Our goal is to provide a recipe for practitioners to use when testing hypotheses about how specific properties of pretraining documents give rise to model behaviors, allowing for flexibility in the scale of the intervention (how much data is changed) and what portion of the training process is studied.

We demonstrate the utility of our recipe through case studies on *factual knowledge acquisition*. LMs have been established as sources of factual information that can be queried with natural language (Petroni et al., 2019; Roberts et al., 2020), but the ways in which pretraining data gives rise to their knowledge capabilities are still un-

der study. We perform two case studies to further this exploration: the first relating term cooccurrence to fact triple completion, and the second using information retrieval methods to study more complex knowledge queries. Both aim to improve our understanding of knowledge acquisition through intervening on and retraining over pretraining data, complementing previous observational studies while also showcasing our recipe’s utility for different data sources and hypotheses.

For our first analysis, we operationalize knowledge acquisition through ⟨subject, relation, object⟩ fact triplets such as ⟨“France”, “capital city”, “Paris”⟩, considering a fact as having been learned when a model can identify the object given the subject and relation, i.e., correctly select “Paris” from a list of candidates when prompted with “The capital city of France is”. We use our recipe to study the hypothesis that fact learning is related to subject-object *cooccurrence* in pretraining documents. Cooccurrence is widely used as a source of distant supervision for aligning facts to text (Mintz et al., 2009; Augenstein et al., 2016; El-sahar et al., 2018) and has a well-established relationship to LM behavior through *observational* analyses on the scale of entire pretraining datasets and fully-pretrained models (Razeghi et al., 2022; Kandpal et al., 2023; Elazar et al., 2022; Kang and Choi, 2023; Zhang et al., 2024; Mallen et al., 2023; Merullo et al., 2025; Wang et al., 2025; Verma et al., 2024; Seshadri et al., 2024). We also apply our recipe to modern multiple-choice question answering (MCQA) benchmarks used to evaluate LM factual knowledge (MMLU, Hendrycks et al., 2021; OpenBookQA, Mihaylov et al., 2018; SciQ, Welbl et al., 2017), exploring another way of expressing and querying for factual knowledge that requires moving beyond cooccurrence and serves to demonstrate the flexibility of our recipe. Our results demonstrate the potential of our recipe to study how pretraining documents give rise to model behavior through direct intervention, complementing post-hoc observational analysis.

2 Overview of Recipe

2.1 Components

We begin with the components of our interventional analysis recipe for studying the relationship between pretraining² data and model behavior:

²We focus only on pretraining in this work, but a similar recipe could be applied to other stages of training as well.

- A pretrained LM $\mathcal{M}$. We assume access to a set of *checkpoints* $\mathcal{M} = \{\mathcal{M}_1, \dots, \mathcal{M}_T\}$, where each checkpoint $\mathcal{M}_s$ consists of model weights at intermediate step s during pretraining and $\mathcal{M}_T = \mathcal{M}$ is the final pretrained model. Since we wish to retrain from intermediate checkpoints, we also assume access to the optimizer state at each checkpoint.
- The pretraining dataset $\mathcal{D} = \{d_1, \dots, d_N\}$ used to train $\mathcal{M}$, where each d_i is a sequence of tokens that form a document.³ We assume the dataset can be partitioned into T *data batches* $\mathcal{D}_1, \dots, \mathcal{D}_T$, such that $\mathcal{D}_s$ is the ordered sequence of tokens seen between checkpoints $\mathcal{M}_{s-1}$ and $\mathcal{M}_s$ during pretraining.
- A task or skill of interest represented by an evaluation dataset $\mathcal{E} = \{(x_1, y_1), \dots, (x_n, y_n)\}$ of *test items*, where each item (x_i, y_i) is an input and its corresponding correct output.
- An *evaluation metric* $\mathcal{F}(\mathcal{M}_s, (x_i, y_i))$ that applies a checkpoint $\mathcal{M}_s$ to an item (x_i, y_i) to quantify model behavior on that item (e.g., correctness at predicting y_i given x_i , log-probability of y_i given x_i , etc.).

Given $\mathcal{D}$ and a full specification of the LM training procedure, we can replicate pretraining from a model checkpoint $\mathcal{M}_s$ to $\mathcal{M}_{s+k}$ for any two checkpoints indexed s and $s+k$, by retraining over the ordered data batches $\mathcal{D}_{s+1}, \dots, \mathcal{D}_{s+k}$. We can also make targeted interventions on those data batches and re-run pretraining to generate a counterfactual model checkpoint $\mathcal{M}_{s+k}^{\text{int}}$, which we propose doing in our interventional analysis recipe.

2.2 Designing an Experiment

Suppose we are interested in how pretraining data affects the behavior of our model $\mathcal{M}$ at some timestep t , as measured by applying our metric $\mathcal{F}$ to $\mathcal{M}_t$ on some target subset of test items $\mathcal{E}^{\text{target}} \subseteq \mathcal{E}$. We have some hypothesis about how the relationship between documents in $\mathcal{D}_t$ ⁴ and items in $\mathcal{E}^{\text{target}}$ cause this behavior, and we wish to intervene on $\mathcal{D}_t$ to test our hypothesis. Our interventional analysis recipe can be broken down into

³Note that each d_i does not have to be a complete document from the original text data, as documents can be split and concatenated together to fit into the maximum sequence length window when training the LM.

⁴Here we consider only data seen since the previous model checkpoint, but in practice the partitioning of $\mathcal{D}$ into data batches can be defined in such a way that $\mathcal{D}_t$ spans multiple checkpoints.

three stages:

- Selecting a subset $\mathcal{E}^{\text{target}} \subseteq \mathcal{E}$ of target evaluation items on which our LM exhibits the behavior at timestep t , and corresponding data batch $\mathcal{D}_t$ to intervene on (§2.3);
- Identifying a set of relevant documents in $\mathcal{D}_t$ by matching to target items from $\mathcal{E}^{\text{target}}$ (§2.4);
- Intervening by modifying relevant documents in $\mathcal{D}_t$ to construct intervention data batch $\mathcal{D}_t^{\text{int}}$ and retraining over it starting from $\mathcal{M}_{t-1}$ to yield intervention checkpoint $\mathcal{M}_t^{\text{int}}$ (§2.5).

Finally, we compare $\mathcal{M}_t$ and $\mathcal{M}_t^{\text{int}}$ by applying $\mathcal{F}$ to measure their behavior on $\mathcal{E}^{\text{target}}$, determining if the intervention confirms our hypothesis.

There are technical decisions to be made at each stage, and these will depend on the specific model behaviors one wishes to study, the hypotheses, and the computational budget. We go into detail for each of these stages below, with examples of how they can be implemented. Note that we do not provide an exhaustive overview of how to implement each stage of the analysis or anticipate all possible interventions. We believe creative choices will be essential in future studies, and offer a starting point of various options and their tradeoffs. Exploration of other ways of implementing each stage will lead to new instantiations of this recipe that study a wider range of questions.

2.3 Stage 1: Selecting Items and a Data Batch

Given an hypothesis about how data influences some model behavior of interest, we need to first select items from our evaluation dataset $\mathcal{E}$ on which our model $\mathcal{M}$ exhibits that behavior. For example, if we are interested in factual knowledge acquisition, we might select questions that our model answers correctly, whereas if we are interested in forgetting behavior we might select items that the model answers incorrectly at a given checkpoint after being correct for some number of previous checkpoints.

Formally, we wish to select a *target subset* $\mathcal{E}^{\text{target}}$ of items from our evaluation dataset $\mathcal{E}$ (i.e., $\mathcal{E}^{\text{target}} \subseteq \mathcal{E}$). Consider a function of each item, all model checkpoints, and a particular timestep of the form $f_{\text{sel}}((x_i, y_i), \mathcal{M}, s) \in \{0, 1\}$, which is 1 when our model exhibits the behavior of interest on item (x_i, y_i) at step s , and 0 otherwise (note that this is a function of all model checkpoints $\mathcal{M}$ since we may be interested in behavior that involves multiple checkpoints, such as the for-

getting example above). Then we define the target subset $\mathcal{E}_s^{\text{target}} \subseteq \mathcal{E}$ of items at a given step s as

$$\mathcal{E}_s^{\text{target}} = \{(x_i, y_i) \in \mathcal{E} \mid f_{\text{sel}}((x_i, y_i), \mathcal{M}, s) = 1\}$$

Since we have different target subsets $\mathcal{E}_s^{\text{target}}$ for each step s , we need to choose a step to intervene on. This can be motivated by different considerations – the stage of pretraining one is interested in, properties of items in $\mathcal{E}_s^{\text{target}}$, etc. One goal may be for $\mathcal{E}^{\text{target}}$ to contain as many items as possible, as a larger sample size improves our ability to measure the true effect of our intervention (Card et al., 2020). This may motivate choosing a target subset and corresponding timestep t as the subset $\mathcal{E}_s^{\text{target}}$ of the largest size

$$t = \underset{s \in [1, \dots, T]}{\operatorname{argmax}} \quad |\mathcal{E}_s^{\text{target}}| \quad \mathcal{E}^{\text{target}} = \mathcal{E}_t^{\text{target}}$$

There are also many ways to design $f_{\text{sel}}(\cdot)$ to select items, and this choice should be guided by the behavior under study. For example, Chang et al. (2024) examine changes in log-probability of target spans in text probes to study factual knowledge acquisition, and Porada et al. (2022) do the same for commonsense knowledge triples. Selection could also be based on intermediate layer activations: Kim et al. (2025) track the entropy of activations in feedforward layers and relate it to fact learning and forgetting.

Existing work typically uses synthetic data or all items in a dataset rather than just ones on which the model exhibits a certain behavior, but we propose selecting a target subset since not all items in a benchmark may be relevant to the hypothesis (e.g., for fact learning, not all items may be learned at a given checkpoint). We believe properly selecting a target set of items is an important consideration when performing these kinds of analyses, and so we include it in our recipe for completeness.

2.4 Stage 2: Identifying Relevant Documents

Given a target subset of items $\mathcal{E}^{\text{target}}$ on which our model exhibits the behavior of interest and intervention timestep t , we next identify a subset of pretraining documents within the data batch $\mathcal{D}_t$ that we hypothesize to affect the behavior on the target items. The matching process should also be motivated by the original hypothesis, i.e., what relationship between documents and items is believed to be driving model behavior.

In practice, it may be hard to identify documents that we believe contribute to model behavior on our target set, so we approximate with some heuristic for matching documents to items. For example, if our hypothesis is that pretraining documents which contain multiple terms that also appear in an item lead to the model making a correct prediction on that item, we should match the item to documents with some lexical overlap (e.g., term cooccurrence). Suppose we define a matching function $g_{\text{match}}(d, (x_i, y_i)) \in \{0, 1\}$, which has value 1 if document d matches item (x_i, y_i) , and 0 otherwise.⁵ We can then define a function that matches a document to our entire target subset of items $\mathcal{E}^{\text{target}}$ as

$$f_{\text{match}}(d, \mathcal{E}^{\text{target}}) = \bigvee_{(x_i, y_i) \in \mathcal{E}^{\text{target}}} g_{\text{match}}(d, (x_i, y_i)),$$

i.e., we consider a document d to be matching if it matches *any* item in our target group $\mathcal{E}^{\text{target}}$. We will use this function to identify documents that we wish to modify to perform our intervention.

There are several considerations when choosing a method of matching documents to items, including tradeoffs in how effective different methods are at identifying target documents. While term cooccurrence can be used as an effective proxy for expression of a relation between entities (Elsahar et al., 2018), spurious matches are possible (e.g., documents where “Paris” and “France” cooccur without expressing the relation “The capital city of France is Paris”). Similarly, Yauney et al. (2023) find that item-document similarity measured using n-gram distributions or pretrained model embeddings does not correlate with performance across a range of tasks, though without performing any interventions. Our recipe provides a way of testing the hypotheses that motivate using each of these methods to identify relevant documents.

Selecting a data batch to intervene on and even defining how much data constitutes a “data batch” in one’s own analysis also requires careful consideration. While frequent checkpoints can lead to data batches with several million documents and enable a wide range of possible matching methods, applying our recipe to large batches or entire pretraining datasets with billions of documents

⁵This formulation can be general – if we instead have a method that assigns a real-valued similarity score to each document-item pair quantifying their relatedness, we can threshold this score to produce an indicator function $g_{\text{match}}(\cdot)$ as described.

would render all but the simplest of methods (e.g., searching for terms) computationally intractable. Thus, potential computational constraints may determine the scale of the intervention.

2.5 Stage 3: Intervening

Our final stage performs the intervention by modifying the documents in $\mathcal{D}_t$ that match items in $\mathcal{E}^{\text{target}}$ and retraining over the modified $\mathcal{D}_t$ to observe the effects. Our intervention should also be motivated by the hypothesized relationship between documents in $\mathcal{D}_t$ and items in $\mathcal{E}^{\text{target}}$ such that it alters (e.g., removes) this relationship. As an example, if we believe that a model correctly answers “What is the capital of France?” with “Paris” because it was trained on documents where “Paris” and “France” cooccur, it is sufficient to intervene on documents in any way that removes this cooccurrence. This can be as small an intervention as editing a few tokens, or as large as replacing the entire document with one in which “Paris” and “France” do not cooccur.

We define an intervention function $g_{\text{int}}(\cdot)$ that takes in a matched document, the set of target items, and the remaining pretraining data and returns a modified document $d' = g_{\text{int}}(d, \mathcal{E}^{\text{target}}, \mathcal{D})$. We include $\mathcal{E}^{\text{target}}$ and $\mathcal{D}$ as arguments to the intervention function since we may want to make modifications that make use of properties of items (e.g., salient terms that occur in x_i or y_i) or replace d with other documents in $\mathcal{D}$. Since we only want to intervene on documents which match items in our target set $\mathcal{E}^{\text{target}}$, we define a function

$$f_{\text{int}}(d) = \begin{cases} g_{\text{int}}(d, \mathcal{E}^{\text{target}}, \mathcal{D}) & \text{if } f_{\text{match}}(d, \mathcal{E}^{\text{target}}) = 1 \\ d & \text{if } f_{\text{match}}(d, \mathcal{E}^{\text{target}}) = 0 \end{cases}$$

Then we construct our intervention data batch $\mathcal{D}_t^{\text{int}}$ from our original data batch $\mathcal{D}_t$ as $\mathcal{D}_t^{\text{int}} = \{f_{\text{int}}(d) \mid d \in \mathcal{D}_t\}$. Finally, we re-run pretraining over the intervention data. We initialize with model checkpoint $\mathcal{M}_{t-1}$ (and its optimizer state) and train over $\mathcal{D}_t^{\text{int}}$ instead of $\mathcal{D}_t$, resulting in an intervention checkpoint $\mathcal{M}_t^{\text{int}}$. We then apply our evaluation metric $\mathcal{F}$ to items in $\mathcal{E}^{\text{target}}$ using both $\mathcal{M}_t$ and $\mathcal{M}_t^{\text{int}}$ separately to compare their behavior and determine if the effect of our intervention is consistent with our hypothesis. The implementation depends on the evaluation metric. For instance, we can compare $\mathcal{F}(\mathcal{M}_t, (x_i, y_i))$ to $\mathcal{F}(\mathcal{M}_t^{\text{int}}, (x_i, y_i))$ for each $(x_i, y_i) \in \mathcal{E}^{\text{target}}$ (e.g., change in log-probability of y_i given x_i), or we

can aggregate $\mathcal{F}(\mathcal{M}_t, (x_i, y_i))$ over all $(x_i, y_i) \in \mathcal{E}^{\text{target}}$ (and similarly for $\mathcal{M}_t^{\text{int}}$) and compare the aggregated values (e.g., accuracy on $\mathcal{E}^{\text{target}}$).

The intervention function $g_{\text{int}}(\cdot)$ can vary in the degree to which it alters the pretraining data, depending on the hypothesis being studied. [Biderman et al. \(2023\)](#) perform a token-level intervention that swaps masculine and feminine pronouns late in pretraining to observe the effects on gender bias, while [Wei et al. \(2021\)](#) add and remove small subsets of documents to see how verb frequencies affect subject-verb agreement performance. At the more extreme end, [Kandpal et al. \(2023\)](#) remove all documents containing cooccurrences of terms from a subset of factual questions, ablating 30% of the pretraining dataset in total before training a counterfactual model from scratch.

Different interventions come with their own pros and cons: token-level interventions enable studying more precise data-behavior relationships but may be too small to have an effect, while removing large sets of documents is more likely to alter behavior but introduces confounding factors like altered token frequencies, domain shift, etc. All of these considerations should be taken into account when designing one’s own document intervention procedure.

In summary, instantiating our recipe requires choosing an item selection function f_{sel} , a document matching function g_{match} , and an intervention function g_{int} .

3 Experimental Setup

We demonstrate our recipe through case studies where we intervene on pretraining documents to study their effect on knowledge acquisition. Our first study relates term cooccurrences to relational fact learning. In the second study, we experiment with multiple-choice question answering (MCQA) datasets MMLU, OpenBookQA, and SciQ to study knowledge acquisition (§5). For all experiments, we use the same model, evaluation setup, method of selecting items, and general intervention procedure, detailed below.

3.1 Model

We use the OLMo models ([Groeneveld et al., 2024](#); [Walsh et al., 2025](#)) for our experiments, as they include all the necessary components for

our recipe as listed in §2.1.⁶ Specifically, we use OLMo 2 1B and 7B with associated pretraining data, OLMo 2 Mix 1124 (Walsh et al., 2025). For the 7B model we use the second and third checkpoints (at 2,000 and 3,000 steps, respectively). We create additional checkpoints at steps 2,150 and 3,150 (retraining over the original ordered data starting from steps 2,000 and 3,000, respectively) to perform interventions over smaller data batches. For the 1B model, we retrain the model from scratch for 10,000 steps on the original ordered data batches, checkpointing every 500 steps to yield more granular checkpoints.

3.2 Evaluation

Key components of our analysis require evaluating checkpoints on the items in each dataset, and we use OLMES to perform these evaluations (Gu et al., 2025). The OLMES framework outlines a variety of choices for prompting the model and ranking answer choices to standardize evaluations for MCQA tasks, which fits all of our datasets.

Following the OLMES notation, we use the cloze/completion task formulation with a QA prompt as input (e.g., “Question: The capital city of France is \nAnswer:”). We rank the LM probabilities of candidate answer choices with different normalization strategies (detailed in the corresponding sections), labeling items as correct if the correct response is ranked highest and using this to calculate accuracy as our evaluation metric $\mathcal{F}$. We use zero-shot prompting for all datasets.

3.3 Selecting Items

We perform two kinds of interventions for a particular data batch: (1) replacing documents that match items learned at a particular checkpoint with unrelated documents in order to *suppress learning*; and (2) replacing unrelated documents with documents that match items learned at a later checkpoint to *promote learning* earlier.

To do so, we first identify at which checkpoints certain facts (represented by items in our evaluation dataset) are first and permanently “learned”. We define the checkpoint at which a fact is first permanently learned as the one where its item is first answered correctly, as long as it also remains correct at all later checkpoints that we evaluate on. We evaluate on the first 30 checkpoints (30K

steps) for OLMo 2 7B and the first 20 checkpoints (10K steps) for OLMo 2 1B.

The early stages of pretraining are where the OLMo models exhibit the most improvement in performance for all datasets. We also find that the early checkpoints are where most items are learned (by our definition), with at least 80% of learned items being labeled correctly by step 4,000 for the 1B model and step 3,000 for the 7B model (see Appendix Fig. 7). For items learned at a particular checkpoint, we identify the subset that have some number of matching documents in the data batch immediately preceding that checkpoint and use those as our target subset $\mathcal{E}^{\text{target}}$.

3.4 Intervening

To perform our intervention, we identify a pair of consecutive model checkpoints $\mathcal{M}_t$ and $\mathcal{M}_{t+1}$ where we either want to *promote learning* of items learned at $\mathcal{M}_{t+1}$ (but not yet learned at $\mathcal{M}_t$) or *suppress learning* of items learned at $\mathcal{M}_t$. In either setting, we intervene on the data batch $\mathcal{D}_t$ just prior to $\mathcal{M}_t$, with details on each intervention described separately for each experiment. In general, we attempt to *suppress learning* by replacing matching documents in $\mathcal{D}_t$ with unrelated ones from $\mathcal{D}_{t+1}$, and we *promote learning* by moving matching documents from $\mathcal{D}_{t+1}$ into $\mathcal{D}_t$ (replacing unrelated documents to do so). Specific details on this document-swapping operation can be found in Appendix §A.1.

After constructing our intervention data batch $\mathcal{D}_t^{\text{int}}$, we take $\mathcal{M}_{t-1}$ and continue training up to the next checkpoint using the same optimizer state, learning rate, and learning rate schedule as the original OLMo 2 pretraining run at that point. Other than the intervened documents, the model will see the same documents in the same order as the original model that was trained from $\mathcal{M}_{t-1}$ to $\mathcal{M}_t$, allowing us to construct an intervention checkpoint $\mathcal{M}_t^{\text{int}}$ that exhibits the effects of only our intervention on the original data batch $\mathcal{D}_t$.

Interestingly, we observe different results even with repeated runs of the exact same portion of pretraining (i.e., the same initial state and ordered data batches). We attribute this to irreducible noise arising from distributing training across GPUs. We repeat training runs over each intervention data batch as well as the original data batch to take this variation into account, reporting the mean and standard deviation across runs for each condition.

⁶The availability of such resources is essential for performing the kinds of analyses we describe, underscoring the importance of fully open LMs for studying pretraining.

4 Term Cooccurrence and Relation Fact Learning

In the first study, we test the hypothesis that the presence of term cooccurrences – for example, the cooccurrence of “Paris” and “France” given the factual relation “Paris is the capital of France” – in the data batch immediately prior to a fact being learned affects whether it is learned at all. Existing work has studied the relationship between term cooccurrence and factual knowledge acquisition either through observational data (Razeghi et al., 2022; Kandpal et al., 2023; Elazar et al., 2022; Kang and Choi, 2023; Zhang et al., 2024) or by pretraining from scratch on synthetic data (Kassner et al., 2020) or modified pretraining datasets (Wei et al., 2021; Kandpal et al., 2023). In contrast to past work, which primarily analyzes the *final* pretrained model, we study the intermediate checkpoints where these facts are *first* learned, by making specific, targeted interventions on the data.

4.1 Experiment Setup

Dataset We use ParaRel⁷ as our evaluation dataset $\mathcal{E}$ (Elazar et al., 2021), which consists of Wiki-Data facts formatted as a relation between two entities – a subject and an object – allowing us to use subject-object cooccurrence as a heuristic to identify matching documents. Each relation is also associated with various templates that express it in natural language, and we use a single template per relation when querying our LM. We use the 12 relations where both models show better than chance performance at the intervention checkpoint, with a subset shown in Table 1 as an example (full details in Appendix Table 6). When evaluating with the OLMES framework, we use no normalization when ranking candidate answer choices, as we found this gave the best performance.

Counting (co)occurrences We count occurrences of entities from ParaRel items mentioned in a batch of pretraining data using WIMBD⁸ (Elazar et al., 2024). This allows us to identify the documents entities occur in and their starting and ending character location within those documents. For items where the strings of the subject

and object have partial overlap (e.g., “Microsoft Windows is a product of Microsoft”), we only consider cooccurrences where the character windows of the two terms do not overlap.

Intervening We attempt to suppress and promote learning of items originally learned at two different pretraining steps each for the 1B and 7B OLMo 2 models, retraining over steps 1.5–2K and 2.5–3K for the 1B model and steps 2–2.15K and 3–3.15K for the 7B model. In each case, we attempt to suppress learning of items originally learned at the end of each data batch, and promote learning of items learned at the end of the next batch. To suppress learning, we use both cooccurrence of subject and object entities as well as occurrence of either entity to identify related documents. The replacement of occurrences is a more aggressive intervention to determine a lower bound on performance change. This results in replacing 0.5–4.7% of documents for cooccurrences and 17.3–43.7% for occurrences, depending on the model size and data batch. For promoting learning, we only use cooccurrences to identify relevant documents to move into the previous batch, replacing 1.3–9.9% of documents for any given model size and pretraining step.

In all cases, we initialize from the OLMo 2 checkpoint at the initial step and continue pretraining for 500 (1B) or 150 (7B) steps on the modified data batch, then evaluate the counterfactual checkpoints on our target subset for which we are attempting to suppress or promote learning. We perform 5 repeated training runs for each result.

4.2 Result: Removing Cooccurrences and Occurrences Suppresses Learning

Fig. 2 shows that performance for learned items is reduced when cooccurrences are removed, and reduced even further when removing occurrences. This trend holds for multiple checkpoints and across model scales. We also compare all results to a *majority baseline* that selects the most commonly-occurring object entity for each relation in the ParaRel training set.

Notably, removing all cooccurrences yields higher performance than the majority baseline, with a 30.7–66.5% absolute difference in accuracy depending on the model size and pretraining step. If an LM’s transition from answering an item incorrectly at one checkpoint to correctly at the next was entirely due to cooccurrences in the data

⁷We use a version of ParaRel that was modified for autoregressive language modeling (CoAStL NLP Group, 2024).

⁸Specifically, we use the `wimbd search` command for fast exact string matching.

Relation	Template	Examples
P17	[X] is located in [Y]	Transylvanian Plateau is located in Romania, Puerto Maldonado is located in Peru
P27	[X] is a citizen of [Y]	Gabriel Axel is a citizen of Denmark, Fumihiko Maki is a citizen of Japan
P178	[X] is a product of [Y]	Picasa is a product of Google, iOS is a product of Apple Inc.
P495	[X] was created in [Y]	The Lemon Bucket Orchestra was created in Canada, Suite Habana was created in Cuba

Table 1: A sample of the relations we use from the ParaRel dataset, with example items for each.

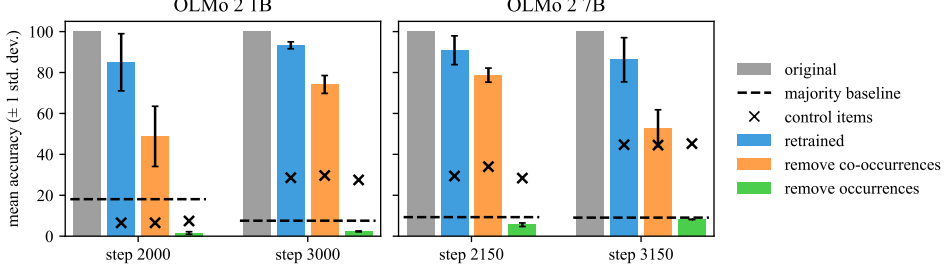


Figure 2: Results for replacing matching documents and re-training checkpoints to *suppress learning* of selected items from ParaRel. Accuracies are shown on the subset of target items. We include results of retraining on the original data batch (blue), training on a modified batch that replaces cooccurrences (orange), and training on a modified batch that (more aggressively) replaces occurrences (green). We also show results on target items (colored bars) and control items (crosses; i.e., all items other than the target group). For all results, we perform 5 repeated training runs and show the mean and one standard deviation across runs.

seen between those two checkpoints, we would expect removing those cooccurrences to drop performance to the majority baseline. As this is not the case, we conclude that while cooccurrences in the most recent data batch encourage learning, they are not responsible for all of it. The remaining difference could be explained by a number of factors – cooccurrences from previous data batches, multi-hop cross-document chains of relations linking the two entities, cooccurrences of different terms with similar model representations, etc. – all of which would be interesting future interventions for further understanding this phenomenon.

In contrast, the more drastic intervention of removing occurrences is sufficient to completely suppress learning for all steps and model scales, although this intervention modifies a much larger fraction of each data batch. This suggests that learning is in fact caused by the documents with entity mentions, although not just by documents that contain cooccurrences. Testing other hypotheses could determine whether the same reduction in performance could be obtained by a smaller intervention.

Finally, we also show the effect of our intervention on a *control group* of all items other than the ones we target for suppressing learning. Our results show that change in control group performance is minimal, indicating that our intervention successfully targets items for which we aim to al-

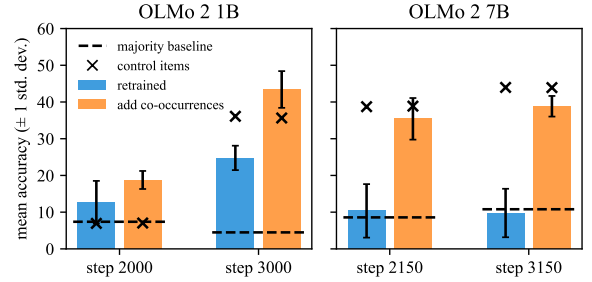


Figure 3: Results for replacing matching documents and retraining to *promote learning*. Replacing documents not containing a cooccurrence with documents that do improves performance for all models and checkpoints.

ter performance.

4.3 Result: Introducing Cooccurrences Promotes Learning

Fig. 3 shows that introducing cooccurrences promotes learning, increasing performance on the target items above their original performance and well above the majority baseline. However, these items were selected because they were all labeled correctly (i.e., “learned”) at the very next checkpoint, and our intervention fails to attain perfect accuracy. As in the previous section, this finding suggests that documents beyond the ones that contain entity cooccurrences in this data batch (or possibly a prior one) also contribute to the model’s acquisition of factual knowledge for these items. As

in the learning-suppression setting, our intervention has minimal impact on control group items.

We perform supplementary experiments varying our method of matching documents to items (Appendix Fig. 8) and our method of intervening on matched documents (Appendix Fig. 9). These experiments show a similar pattern to what is reported above.

5 Relevance and Complex Knowledge Queries

While facts expressing an entity pair relation provide a useful format for studying knowledge acquisition via term cooccurrence, this is not always the way knowledge is expressed. Modern MCQA datasets test LM knowledge through complex questions paired with several possible answers, with questions and answers sometimes containing references to multiple entities or expressing abstract concepts involving no salient entities at all. Since term cooccurrence cannot be easily used to identify relevant documents for these items, we instead use methods from information retrieval (IR). As with our experiments on ParaRel, we believe that information in pretraining documents is contributing to knowledge learning for our MCQA datasets, and we hypothesize that IR methods are an effective way of identifying documents that contribute to knowledge acquisition.

5.1 Experiment Setup

Datasets We use common datasets for evaluating or comparing models (Groeneveld et al., 2024; Walsh et al., 2025; Gu et al., 2025): MMLU (Hendrycks et al., 2021), OpenBookQA (Mihaylov et al., 2018), and SciQ (Welbl et al., 2017). OpenBookQA and SciQ consist of exam-style science questions, while MMLU contains questions from 57 subjects. For all datasets, each item is formatted as a question with a set of 4 multiple-choice answer options (e.g., from OpenBookQA: “What doesn’t eliminate waste? (a) plants (b) mushrooms (c) bacteria (d) robots”). Following OLMES, we use PMI normalization for OpenBookQA and character normalization for MMLU and SciQ (full details in Appendix Table 7).

Retrieval-based matching Since we often cannot easily identify terms from items in these datasets to search for in pretraining documents, we instead use IR methods to match items to

documents. Specifically, we use the keyword-based method BM25 (Roberts et al., 2020) and the embedding-based method Dense Passage Retriever (DPR; Karpukhin et al., 2020). BM25 is a popular method that scores a document’s relevance to a search query based on relative word frequencies,⁹ while DPR uses trained models to compute item and document embeddings and scores a document’s relevance to an item using embedding similarity.¹⁰ We include both the question and the correct answer in the search queries for BM25 and to construct query embeddings using DPR.

Intervening For our MCQA datasets, we similarly perform interventions to either *suppress* or *promote* learning. When using BM25 and DPR to identify matching documents, we use the fact that each method assigns a real value similarity score to each document-item pair to construct our intervention. Specifically, we replace the top- k highest-scoring documents from $\mathcal{D}_t$ the lowest-scoring documents from $\mathcal{D}_{t+1}$, while matching token counts. We match documents greedily, iterating over relevant documents in order of greatest to least number of matching items and finding a document to replace each one. In practice, this results in a lower average relevance score across replaced documents.

We intervene at steps 2K–2.5K for OpenBookQA and SciQ and steps 1K–1.5K for MMLU at the 1B scale, and steps 2K–2.15K for all datasets at the 7B scale; these steps were chosen because they correspond to stages when many items were learned. In all cases, we replace $k = 1K$ documents per item,¹¹ which results in replacing 10.2–11.1% of documents for MMLU, 3.5–6.6% of documents for OpenBookQA, and 10.9–21.3% of documents for SciQ, depending on the method and model scale. We perform 3 repeated training runs for each setting.

5.2 Result: Removing Relevant Documents Suppresses Learning

Results for *suppressing learning* are in Table 2, with all interventions leading to a decrease in average accuracy. Using DPR scores seems to have less of an effect than using BM25 scores across datasets and model scales, even when matching

⁹We use the `bm25s` Python package.

¹⁰We use the multiset DPR encoders from Facebook (Wolf et al., 2020).

¹¹Exploratory experiments suggested that increasing to 2K documents per item changed the results very little.

Model	Intervention	MMLU		OpenBookQA		SciQ	
		acc.	Δ acc.	acc.	Δ acc.	acc.	Δ acc.
OLMo 2 1B	retrained	89.8 _{0.8}	—	88.7 _{2.0}	—	89.8 _{1.4}	—
	replace high-scoring BM25 docs	76.0 _{0.6}	−13.8	70.2 _{3.0}	−18.5	48.3 _{2.3}	−41.5
	replace high-scoring DPR docs	86.0 _{2.8}	−3.8	78.0 _{2.0}	−10.7	59.4 _{5.7}	−30.4
	majority baseline	26.9	−62.9	27.6	−61.1	25.0	−64.8
OLMo 2 7B	retrained	84.8 _{3.7}	—	78.0 _{4.9}	—	80.6 _{2.5}	—
	replace high-scoring BM25 docs	46.8 _{2.1}	−38.0	65.3 _{4.1}	−12.7	50.6 _{2.4}	−30.0
	replace high-scoring DPR docs	57.1 _{6.5}	−27.7	70.0 _{1.6}	−8.0	55.0 _{1.6}	−25.6
	majority baseline	26.9	−57.9	27.6	−50.4	25.0	−55.6

Table 2: Results on *suppressing learning* for MCQA datasets. Results show the mean and standard deviation (subscript) across 3 training runs. Δ acc. is the change in accuracy as compared to the mean accuracy of the retrained model.

Model	Intervention	MMLU		OpenBookQA		SciQ	
		acc.	Δ acc.	acc.	Δ acc.	acc.	Δ acc.
OLMo 2 1B	retrained	21.3 _{0.0}	—	15.2 _{5.4}	—	20.4 _{3.3}	—
	replace high-scoring BM25 docs	29.1 _{4.8}	+7.8	22.7 _{0.0}	+7.5	47.1 _{0.7}	+26.7
	replace high-scoring DPR docs	22.0 _{1.3}	+0.7	13.6 _{2.3}	−1.6	34.3 _{7.2}	+13.9
	majority baseline	26.9	+5.6	27.6	+12.4	25.0	+4.6
OLMo 2 7B	retrained	13.6 _{3.6}	—	14.9 _{3.0}	—	11.9 _{3.7}	—
	replace high-scoring BM25 docs	53.7 _{4.6}	+40.1	34.5 _{7.0}	+19.6	59.4 _{2.0}	+47.5
	replace high-scoring DPR docs	31.9 _{2.6}	+18.3	25.3 _{5.2}	+10.4	49.6 _{1.1}	+37.7
	majority baseline	26.9	+13.3	27.6	+12.7	25.0	+13.1

Table 3: Results on *promoting learning* for MCQA datasets. Results show the mean and standard deviation (subscript) across 3 training runs. Δ acc. is the change in accuracy as compared to the mean accuracy of the retrained model.

the number of replaced documents per item. Similar to ParaRel, the performance after our intervention is still well above the baseline for all datasets, suggesting that the documents identified by the IR methods are still not entirely responsible for knowledge learning.

5.3 Result: Adding Relevant Documents Promotes Learning

Table 3 shows results for *promoting learning*, where similarly to suppressing learning all of our interventions cause a change in average accuracy. The change in performance is again larger when using BM25 scores rather than DPR scores across all datasets and model scales. In this setting, there is a consistently larger absolute increase in accuracy at the 7B model scale for all datasets, a pattern which was not present when suppressing learning. Given existing results showing that performance improves more for larger models with methods like instruction fine-tuning (Chung et al., 2024) or parameter-efficient techniques like prompt-tuning (Lester et al., 2021), this effect may

be the consequence of larger models being more amenable to performance improvement.

We additionally analyze the relevance of documents identified by each of our matching methods, with full details in Appendix A.2. Overall, we find that our matching methods often do not identify documents with enough information to answer each question, potentially explaining why our interventions were not entirely effective and leaving room for more precise matching methods to be explored in future work.

6 Related Work

The most closely related work was highlighted earlier; here we note additional strands of interest.

Training data attribution Our work is related to quantifying the impact of a training example on a model’s prediction (Hammoudeh and Lowd, 2022). Among the most popular methods for doing so are gradient-based techniques such as influence functions (Koh and Liang, 2017), which have been applied to LM fine-tuning data (Han

et al., 2020) as well as pretraining data (Grosse et al., 2023), with the latter requiring approximations due to high computational cost. These approaches have their own limitations, as they sometimes perform comparably to cheaper alternatives like BM25 at identifying relevant documents (Akyurek et al., 2022) and may assign high influence to documents which do not express the information (Chang et al., 2025). While often used to explain a model’s predictions rather than test hypotheses about how data causes behavior, influence functions could be integrated into our recipe for identifying target documents to intervene on.

Continued training In contrast to our method of intervening at intermediate steps, many other works continue training from the final checkpoint on select data to study behavior or improve LMs. Razeghi et al. (2023) continue pretraining on documents selected via embedding similarity or regular expression matching to improve math reasoning. Embedding or gradient similarity has also been used to select data for continued training to adapt LMs to domains and tasks (Gururangan et al., 2020), boost performance on certain tasks (Han and Tsvetkov, 2022), improve in-context learning ability (Han et al., 2023), or improve instruction-following (Iverson et al., 2023; Xia et al., 2024). Notably, the last set of approaches focus on boosting performance rather than understanding LM behavior.

Other related work In recent work, Gottesman et al. (2025) released a suite of models and their pretraining data with entities annotated to promote studying how data affects knowledge acquisition. While they do not perform the kinds of interventions we explore here, the artifacts they provide enable similar analyses. Bordt and Pawelczyk (2025) retrain OLMo 2 1B from scratch on a modified subset of its pretraining data to simultaneously replicate multiple past experiments, including those that study knowledge acquisition. Each experiment they perform can be considered an instantiation of our proposed recipe for studying a particular model behavior.

7 Discussion and Conclusions

We have introduced a flexible recipe for interventional studies of the effects of pretraining data on LM behavior, formalizing each stage to enable systematic exploration. Our recipe highlights the

importance of targeted interventions built around specific test items and studying changes at intermediate steps, not merely the final model.

Our results show that simple interventions that add or remove documents relevant to the fact (either using term cooccurrence or information retrieval), have a consistent effect on promoting and suppressing learning of facts. These findings support the position that exposure to a textual expression of a fact in training helps a model learn to correctly report it at test time. However, adding relevant documents does not guarantee learning, and removing them does not always prevent learning. More study is required to fully characterize the process by which language models acquire facts, and we believe the methodological tools given by our recipe will aid those future efforts.

8 Limitations

As we designed our recipe to be general enough to test a variety of hypotheses, we focus here on the limitations of our case studies on knowledge acquisition in LMs. Our interventions attempt to promote and suppress learning of facts but are only partially effective, for many potential reasons. While cooccurrence and information retrieval are well-established methods of identifying potentially relevant documents, they are ultimately based on heuristics and it is doubtful that they identify *all* documents that express the fact in a particular test item. LMs may also acquire knowledge through connections across multiple documents, such that matching individual documents is insufficient. The notion of a “document” could be expanded to account for this when matching items to documents. Finally, while our recipe is general enough to intervene on all data since the start of pretraining, we avoid doing so due to computational limitations. The remaining performance gap in our experiments may be due to information in documents from past data batches that we did not intervene on, and a more involved study could apply our recipe to test this hypothesis.

Acknowledgments

This research is supported by the National Artificial Intelligence Research Resource (NAIRR) Pilot and the Anvil supercomputer supported by the National Science Foundation (award NSF-OAC 2005632) at Purdue University, as well as NVIDIA resources provided through NAIRR.

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A Appendix

A.1 Swapping Documents Between Batches

All of our interventions involve taking documents from another data batch and using them to replace documents in the data batch that we wish to intervene on. We replace documents while matching total token counts when possible to minimize altering the data batch. For over 96% of documents, the document that replaces it has exactly the same number of tokens. In a few cases, the document being swapped in is too long by a few tokens, which requires shifting the remaining tokens in the sequence to the right and truncating at the end. Fig. 4 shows a toy example of how this document-swapping operation is performed with token IDs across two batches of tokens.

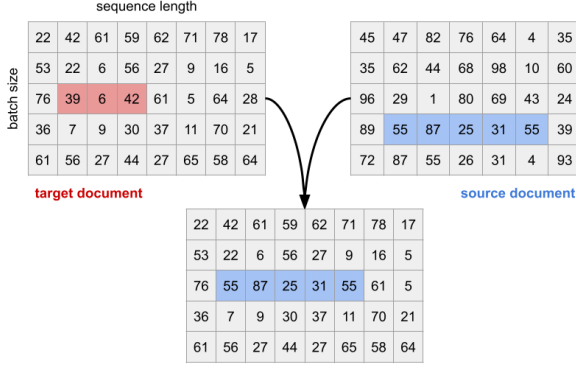


Figure 4: Toy example that shows how token IDs in a batch of tokens are altered when replacing a target document with a source document. Note how the token IDs to the right of the target document are shifted to the right to accommodate the longer source document.

A.2 Determining Document Relevance

While cooccurrence, BM25, and DPR are all reasonable methods of matching documents to items, ultimately they are approximations of our primary objective: identifying documents that express the information of a particular item. We assess the relevance of the documents matched by each of these methods by using GPT-5 (OpenAI, 2025) as a judge to annotate a subset of documents. While using GPT-5 to perform the actual document matching over an entire data batch containing millions of documents is not scalable, using GPT-5 to annotate a subset of documents allows us to analyze the performance of our other more efficient matching methods. We also perform manual annotation on a smaller subset to compare to the GPT-5 annotations.

Setup For each combination of dataset (ParaRel, MMLU, OpenBookQA, and SciQ) and matching method (cooccurrence (ParaRel only), BM25, and DPR), we select the top-20 highest-scoring documents for each intervention group item from our OLMo 2 1B experiments to form a subset for annotation.¹²

For each item-document pair, we provide GPT-5 (gpt-5 in the OpenAI API, accessed September 28th, 2025) with the question, correct answer, and document along with one of three prompts to elicit “yes”/“no” annotation of varying levels of document relevance. We use “correct answer”,

¹²Since cooccurrence does not provide a score for each document, we instead use a random sample of 20 matching documents.

“educated guess”, and “related topic” prompts, which are designed to determine whether the document contains sufficient information to correctly answer the question, contains enough information to make an educated guess about the answer (while not containing the answer), or is topically related to the question despite not containing enough information to answer it, respectively. The specific format of each prompt can be found in Table 4. We set the verbosity to “low” and the reasoning level to “minimal” when querying the OpenAI API.

Results Table 5 shows example items from MMLU with documents matched by BM25 score that were annotated by GPT-5 as being relevant or not, for each of the three annotation prompts. A summary of the GPT-5 annotation results can be found in Fig. 5, where we plot the percentage of item-document pairs for which GPT-5 responded “yes” when asked to judge document relevance given each prompt. Overall, 9.8–61.3% of documents are determined to be relevant according to the “correct answer” prompt, 15.9–59.6% according to the “educated guess” prompt, and 39.8–91.8% according to the “relevant topic” prompt, with percentages varying considerably across datasets. This suggests that while our matching methods can identify documents that are generally relevant to a question, they are less precise when it comes to identifying documents that directly entail the answer and could provide a model with enough information during pretraining to make a correct prediction.

For ParaRel, cooccurrence performs comparably to BM25 at identifying relevant documents across prompts, likely because each item has a simple format with the subject and object entity names being the most relevant keywords. DPR performs worse than BM25 across all datasets at identifying relevant documents. These results align with our interventional analysis findings, where using BM25 scores to select documents led to both greater promotion and suppression of learning (depending on the experiment) than using DPR scores.

Human Validation To manually validate the GPT-5 annotations, we further select a random subset of 20 item-document pairs for each {dataset, matching method} pair which were then annotated by one of the authors following the same prompts. The corresponding manual annotation

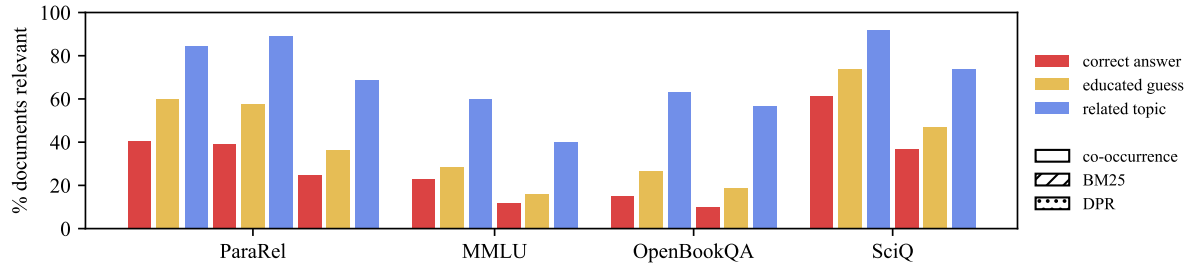


Figure 5: Percentage of documents labeled relevant to their matching item by **GPT-5**, for every dataset, prompt, and matching method.

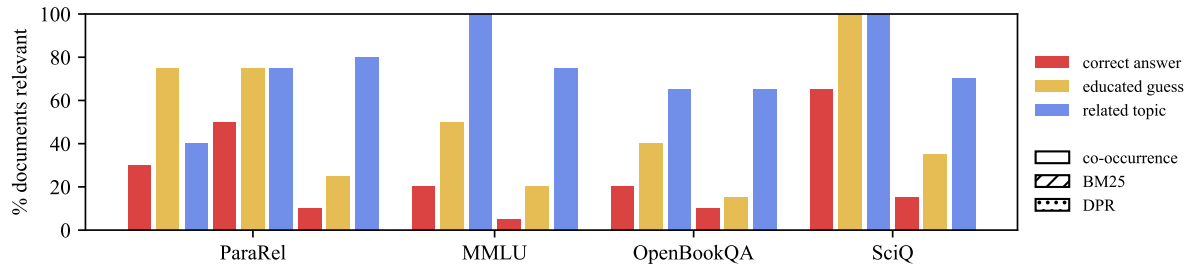


Figure 6: Percentage of documents labeled relevant to their matching item by **manual annotation**, for every dataset, prompt, and matching method.

results can be found in Fig. 6. Generally, the trends are similar to the GPT-5 annotations, although the exact percentages are different. We assess human/GPT-5 annotation agreement by computing Cohen’s κ which gives a value of 0.54 overall, indicating a moderate level of agreement.

Prompt Name	Template
correct answer	You will be given a question or question stem and its correct answer, along with a document. Output “yes” if the document contains enough information to correctly answer the question, and “no” otherwise. Only output “yes” or “no”. Question: {question} Answer: {answer} Document: {document}
educated guess	You will be given a question or question stem and its correct answer, along with a document. Output “yes” if the document contains enough information to make an educated guess about the answer, and “no” otherwise. The document does not have to have enough information to be able to correctly answer the question. Only output “yes” or “no”. Question: {question} Answer: {answer} Document: {document}
related topic	You will be given a question or question stem and its correct answer, along with a document. Output “yes” if the document is topically related to the question and answer, and “no” otherwise. The document does not have to have enough information to be able to correctly answer the question. Only output “yes” or “no”. Question: {question} Answer: {answer} Document: {document}

Table 4: Prompt templates used for document relevance annotation with GPT-5.

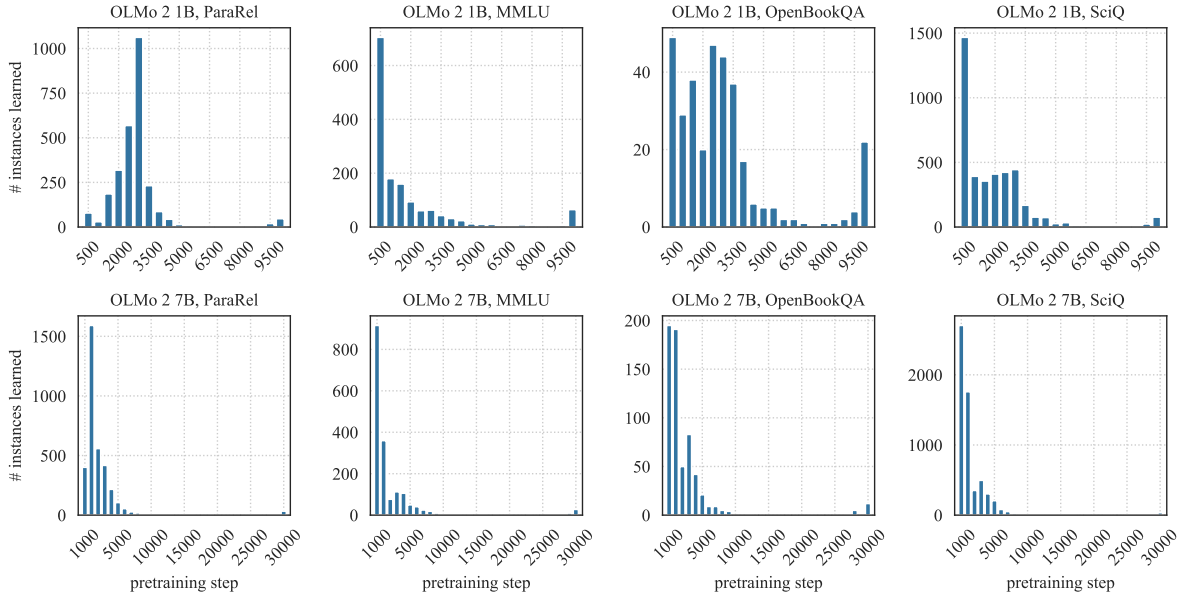


Figure 7: Number of items learned per pretraining step, for each model size and dataset.

Item	Prompt	Document with “yes” annotation	Document with “no” annotation
Question: What do herpetologists study? Answer: Reptiles and amphibians	correct answer	What Classes Do I Need to Take in High School To Prepare for Getting a Zoology Degree? Zoologists study the impact of the environment on animal habitats... Some zoologists will specialize in a species, such as herpetologists, who study reptiles and snakes. Taking science courses while in high school will help you determine in which area of zoology you will specialize... This career may be physically demanding when spending long periods outdoors in rough conditions while researching.	Amphibians vs Reptiles Reptile? Or amphibian? Telling the difference. For most people, these animals are pretty much the same.... Definition Amphibian means “living double lives” on water and land. Reptile means “creeping stealthily under cover of darkness.”... Tetrafauna AquaSafe – The complete water conditioner.
Question: The use of ‘indentured labour’ in the nineteenth century involved: Answer: people being transported to the British colonies and forced to work for one employer under poor conditions	educated guess	History of Guyana Guyana history The recorded history of Guyana can be dated back to 1499, when Alonso de Ojeda’s first expedition arrived from Spain at the Essequibo River... Subsequently, in 1746, the Dutch authorities opened the area near the Demerara River to British immigrants because they were eager to attract more settlers... By 1786 the internal affairs of this Dutch colony were effectively under British control, though two-thirds of the plantation owners were still Dutch. Africans and Indians Africans were enslaved and transported to Guyana as slaves; however, East Indians came as indentured labourers who worked in order to provide for their families back home. By the 1660s, the enslaved	Banner Viadrina Keynote Lecture: Prof. Dr. Andreas Eckert Blurred Frontiers. “Unfree” and “Free” Labour since 1800 in a global perspective In the world of work relations and labour policies, the distinction between “free” and “unfree” labour has been crucial... This lecture critically discusses widespread assumptions which link “unfree” or forced labour to pre-industrial societies and instead argues that there was a close and complex connection between definitions and practices of unfree labor in Europe and in the colonies... The talk will present various historiographical debates related to the issue of unfree/free labour, but will also focus on what historical actors in different parts of the world themselves understood by unfree/free labour.
Question: When you squeeze an air-filled party balloon, you increase its Answer: density	related topic	hydrogen is 0.09 kg/m ³ . Two balloons are inflated to the same size, one filled with air and the other filled with hydrogen. Which balloon experiences the greatest buoyant force? 6) A ten kilogram brick is lowered into a tub of water. A volume equal to two litres of water is displaced. If the density of water is 1g/cm ³ what is the reading of the scale when the brick is fully submerged? Does the brick lose mass when it is placed in water? Activity to measure the buoyant force. Put- Put boats and buoyancy	The Quartet Machine My instrument is called “The Quartet Machine.” It is a member of the string and woodwind families. It is made out of a wooden box, rope, string, beads, plastic bar, rubber bands, balloon, barrette, “squeakies,” and a pencil... The second is you can blow air into the balloon and pinch it near the opening and pull. The third is you can squeeze the “squeakies.” To change the pitch of this instrument you can do one of the following... Finally, to change the dynamics or volume of this instrument you can pluck the strings harder, squeeze the balloon harder or pinch the “squeakies” tighter.

Table 5: A sample of items from MMLU with example documents matched by BM25 score that were given “yes” and “no” annotations by GPT-5, for each of the three annotation prompts.

Relation	# Items	Template	Example
P17	928	[X] is located in [Y]	Mezhdurechensky District is located in Russia
P27	966	[X] is a citizen of [Y]	Marc Lavoie is a citizen of Canada
P36	681	The capital city of [X] is [Y]	The capital city of Greece is Athens
P127	687	[X] is owned by [Y]	Colchester Community Stadium is owned by Colchester
P131	881	[X] can be found in [Y]	Ryerson University can be found in Ontario
P138	604	[X] is named after [Y]	Great Britain is named after Britannia
P176	961	[X] is a product of [Y]	Ferrari 365 is a product of Ferrari
P178	580	[X] is a product of [Y]	Windows Embedded Compact is a product of Microsoft
P276	877	[X] is located in [Y]	1996 Summer Olympics is located in Atlanta
P495	907	[X] was created in [Y]	Lancashire wrestling was created in England
P1376	216	[X] is the capital city of [Y]	Ottawa is the capital city of Canada
P1412	965	[X] communicated in [Y]	Lon Chaney communicated in English

Table 6: All relations we used from ParaRel along with number of items, query template, and an example item for each.

Dataset	Subset(s)	# Items	Example
MMLU	test	14,042	The Rhine River is an example of which type of boundary? (a) geometric (b) artificial (c) natural (d) relict
OpenBookQA	train, validation, test	5,957	What functions without contact between objects? (a) weight (b) friction (c) opposites attracting (d) pressure
SciQ	train, validation, test	13,679	In order to create food, what do photosynthetic protists use? (a) decayed matter (b) thermal energy (c) hydrocarbons (d) light energy

Table 7: Additional details on MMLU, OpenBookQA, and SciQ including subsets used, number of items per dataset, and an example item.

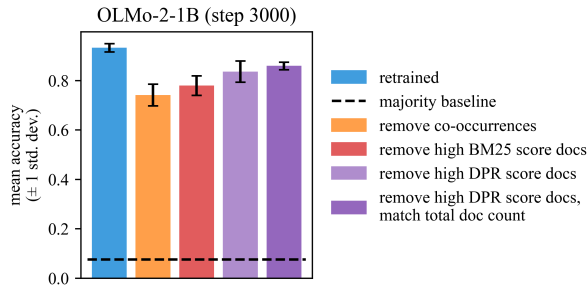


Figure 8: Varying the method of matching documents to items results in different levels of performance reduction on ParaRel items. Selecting relevant documents using BM25 and DPR scores both lead to similar or less reduction in performance compared to selecting based on term cooccurrence alone, when removing the same number of documents per item or in total.

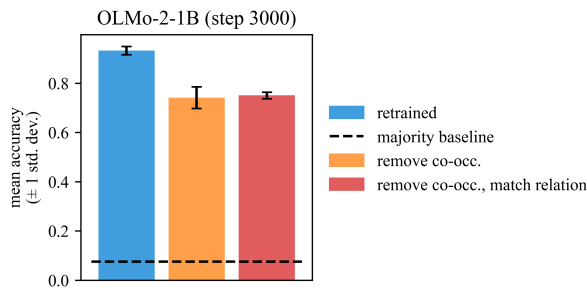


Figure 9: Modifying documents by replacing them with ones expressing the same ParaRel relation type but without having the term cooccurrence has a similar effect to replacing with random unrelated documents, indicating the importance of the specific entity pair being mentioned.